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Nothing Escapes the Pull of a PlayStation 3, Not Even a Black Hole

Scientists scrimp on supercomputer costs but still manage to solve a cosmic quandary with a little help from Sony's PS3.

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How did Sony's PlayStation 3 solve an astrophysics puzzler? Easy: With a pallet full of sleek sable doppelgangers crunching the interstellar math in tandem.

Put that another way, what do you get when you take not one, not four, not eight, but *sixteen* PS3s and chain them together like some crazy video game test center configuration for extremely lucky children? Simple: The relatively affordable power to answer the question "At what speed to vibrating black holes stop vibrating?"

Scientists at The University of Alabama in Huntsville and the University of Massachusetts, Dartmouth, have done just that by harnessing the computing muscle of 16 PS3s running in a homebrew supercluster they've aptly dubbed the "PS3 Gravity Grid." Total cost? Around \$6,000.

That's a deal if you speak supercomputerese. According to Dr. Lior Burko, an assistant physics professor at The University of Alabama, using PS3s instead of renting CPU hours at the National Science Foundation's [TeraGrid](#) or the [Alabama Supercomputing Center](#) was about cost plus run-time flexibility. "If we had rented computing time from a supercomputer center it would have cost us about \$5,000 to run our simulation one time," said Burko. "For this project we ran our simulation several dozens of times to test different parameters and circumstances, so you can see how much that would have cost us."

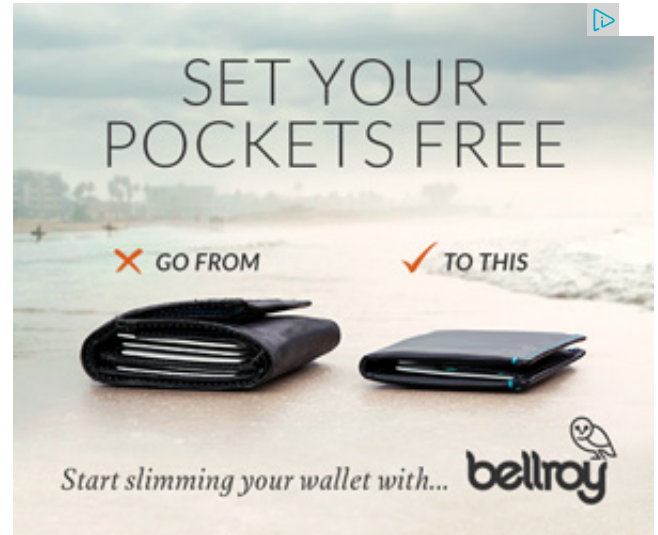
Pairing the 16 PS3s was apparently straightforward enough that University of Massachusetts Dartmouth physics professor Gaurav Khanna, who built the cluster, recently threw together an open source do-it-yourself [PS3Cluster Guide](#) for the budding (not to mention profligate) PS3 supercomputing wannabe. What you need: [Fedora](#) 8, a USB memory stick, a USB keyboard, and a store that actually has 16 PS3s in stock. An actual scientific purpose helps, too.

Speaking of, what's the deal with black holes vibrating? I thought they just *ate* stuff, like in the 1979 Disney movie with the pre-Super-Mario Goomba-robot thing.

Turns out they can actually vibrate after forming...or chomping on something.

"Think of a bell," said Burko. "A bell rings, but eventually it gets quiet. The energy that goes out with the sound waves is energy that the bell is losing. A black hole does exactly that in gravitational waves instead of sound waves. A black hole that is wobbling is emitting gravitational waves. When those vibrations die down you get a quiet black hole."

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Using the PS3 cluster, the physics-on-a-shoestring science team simulated a black hole, "perturbed" it, then measured the point at which it settled back to its "quiet" state.

(Disclaimer: The chances of a PS3 supercluster spawning [a micro black hole](#) that plummets to the center of the planet and slowly devours the earth are probably pretty slim.)

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